GROVASTRA / CORRECTED SCOPE / V2.0

Auto campaigns Developer handoff

React frontend - Node.js backend - MongoDB database

For handloom stores, boutique owners and apparel shops. Prepared 13 October 2026.

Build the auto-campaign system end to end

The assigned developer implements **campaign configuration, customer eligibility, triggers, scheduling, multi-step execution, stop rules, reporting and message dispatch**. A birthday, anniversary or stock event must automatically create and execute the right campaign steps without staff selecting each recipient manually.

THIS DEVELOPER	OTHER DEVELOPER
React auto-campaign screens and settings; Node campaign engine and scheduler; MongoDB campaign records; business rules, edge cases and tests.	WhatsApp account setup, credentials provisioning, account approval, template creation/submission/approval and the authoritative template catalogue.
Select an existing approved template for each campaign step and send it through the provisioned connection.	Provide ready connection details, approved template variants, status changes and authenticated provider events.

What "one template or message" means

Each due campaign step sends **one selected message to one eligible customer**. A campaign may enroll many customers and contain several scheduled steps. This does not reduce the project to a manual single-message sending screen.

Document status

This v2 specification supersedes the earlier sender-focused handoff. WhatsApp is the only delivery channel in this release. Most of the work described here is campaign logic; template approval and account approval are explicitly excluded.

Example lead times, limits and priorities are proposed product defaults and are labelled as such. They are not claims about provider limits or guaranteed sales.

READ FIRST

Reading map and key decisions

Pages	Subject
3-5	Scope, architecture and campaign catalogue
6-8	React screens, campaign settings and message mapping
9-10	Lifecycle, versioning and audience rules
11-16	Detailed industry campaign scenarios
17-20	Scheduling, collision handling, execution and pause/edit behaviour
21-25	MongoDB models, configuration, events, indexes and transactions
26-28	Node APIs, delivery boundary and reporting
29-32	Edge cases and acceptance tests
33-35	Worked example, delivery checklist and references

Release decisions

- Provide preset-driven campaigns with a finite sequence of 1-3 steps. A general visual workflow builder and arbitrary user code are not required.

- Support annual date triggers, explicit event/season dates, domain-event triggers and inactivity scans.

- Use approved-template messages for automated outbound campaigns. Custom free-text editing is not a campaign-template approval bypass.

- Published campaign versions are immutable. Existing enrollments retain their version; current opt-out, stock, order and provider state are always rechecked.

- Use a MongoDB-backed scheduler and durable worker jobs. A MongoDB replica set supports the specified transactions. Redis is not required.

- No bulk WhatsApp request is assumed: fan out safely to one execution per customer per campaign step.

Terms

Campaign: saved rules and steps. **Occurrence:** one birthday year, collection launch, enquiry or other trigger instance. **Enrollment:** one customer participating in that occurrence. **Execution:** one planned message step. **Attempt:** one provider submission for an execution. These are distinct records and metrics.

01 / SCOPE

Ownership and definition of done

Area	Assigned developer implements	Dependency supplied by others
Campaign UX	Preset library, draft editor, review/dry run, activate, pause, history and settings.	Product owners approve proposed defaults.
Triggers and audiences	Birthday/anniversary calculation, seasonal events, event consumers, inactivity rules and customer matching.	CRM, order, stock and appointment data contracts.
Scheduling	Enrollment, local-time planning, durable due jobs, catch-up, expiry and cancellation.	Reliable time and MongoDB deployment.
Business safeguards	Live stock/offer/appointment checks, purchase/reply stops, contact caps and collision priority.	Authoritative domain records; do not build a new POS as part of campaigns.
Message execution	Bind approved template parameters, submit one message per step, dedupe, safe retries and status projection.	Ready connection, approved templates, credentials and provider event feed.
Reporting	Campaign counts, reasons, customer history and explicitly linked orders.	Recorded order data and any available campaign link evidence.
Approvals	Read status and block unusable configuration.	Other developer owns all WhatsApp/account/template approval processes.

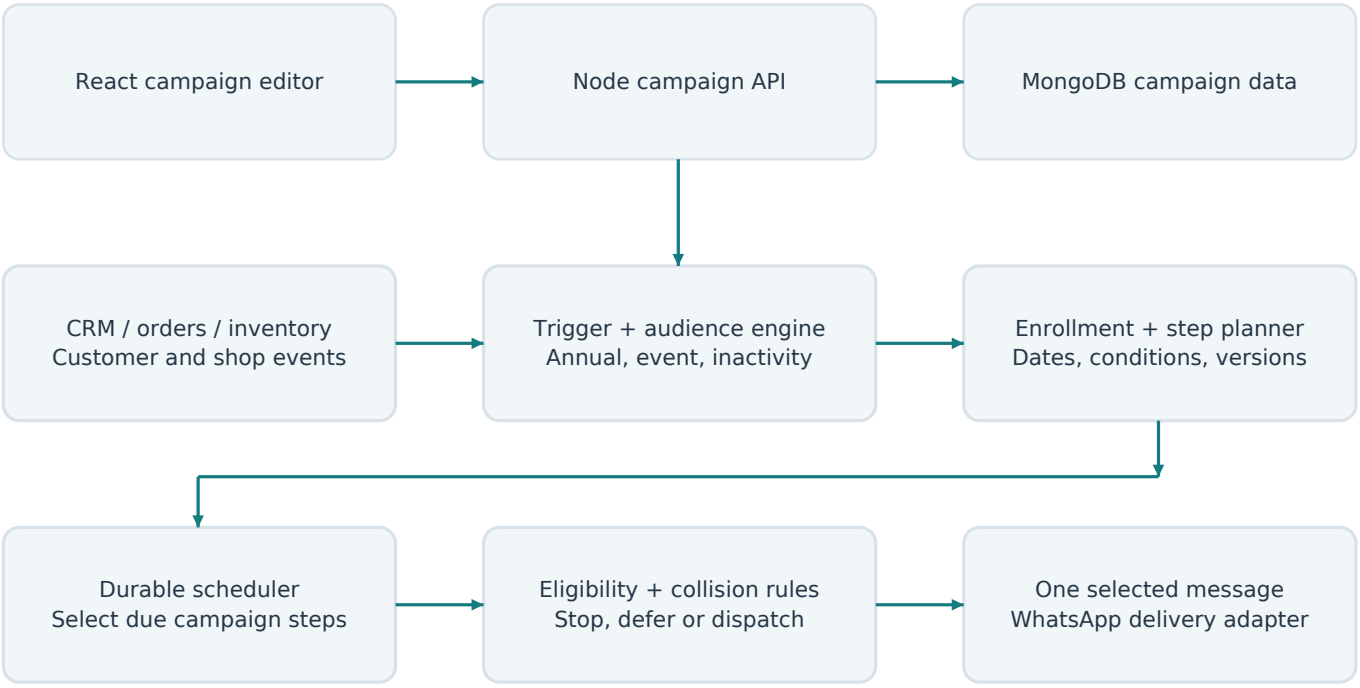
A complete vertical slice

An owner activates a birthday campaign. The engine finds a matching customer, plans a message before the birthday, selects that customer automatically, resolves the saved template, sends when due, and stops later sales follow-ups after a reply or qualifying purchase. The owner can inspect why the customer was selected and what happened.

A working manual Send button alone is not completion. A running scheduler that ignores opt-outs, duplicate events or changed order state is also not completion.

02 / ARCHITECTURE

System architecture and campaign flow



Processes

One Node codebase may expose the campaign API and run separate planner, event-consumer and execution-worker processes. All durable state is in MongoDB. Process timers only wake workers; they do not hold the sole copy of a future schedule.

Business inputs

Consume customer changes, qualified orders, stock publications, appointments, credits, replies and permission changes. Calendar and inactivity scans cover time-based triggers and reconcile missed events. Event replay must be safe.

Delivery boundary

Campaign rules decide **who, when, why and which saved template**. The delivery adapter submits that selected template. The integration developer supplies approval/readiness data; approval operations do not appear in the campaign editor.

03 / CAMPAIGN CATALOGUE

Campaign library and trigger families

Preset	Trigger family	Shop value
Birthday / anniversary	Annual customer date	Occasion shopping and gifts.
Gift / wedding reminder	Customer-declared event date	Help customers prepare in time.
Festival / exhibition	Owner-approved dated event	Seasonal shopping and store visits.
Unresolved enquiry	Enquiry event + elapsed time	Follow up before interest is lost.
Requested item available	Inventory publication + saved request	Contact customers already looking for the item.
Matching collection	Published collection + preferences	Show relevant new arrivals.
Complete the outfit	Qualifying purchase + delay	Relevant blouse/accessory/complement suggestions.
First to second purchase	First qualifying order + delay	Encourage a useful repeat visit.
Win-back	Category-specific inactivity threshold	Invite past customers back with a reason.
VIP / credit / referral	Milestone, real expiry or completed order	Recognition, redemption and referral opportunities.
Fitting / alteration	Appointment or production milestone	Keep boutique customers informed.
Outfit collection / delivery	Actual fulfillment milestone	Reduce missed collections and status questions.
Selected-stock promotion	Owner publishes valid stock offer	Match an offer to suitable customers.

Marketing presets and order-service presets share the engine but have different purposes, templates, stop rules and caps. Do not turn a service update into a promotional message by inserting an offer into a variable.

All presets belong to the model. Implement birthday and anniversary first as the reference slice, then add the remaining families in the delivery order on page 34. No preset may appear as functional in React until its evaluator is implemented.

04 / FRONTEND

React screens and owner journey

Auto campaigns / Birthday shopping

Trigger > Audience > Messages > Sending rules > Review

When
14 days before birthday
10:00, Asia/Kolkata

Who
Opted-in apparel customers
with a valid birthday

Message step 1
Approved birthday template
Map name, shop and offer

Review and activate
Dry-run counts + exclusions
No sends during preview

Screen	Required behaviour
Auto campaigns overview	Preset cards; Draft/Active/Paused/Archived status; next activity; data/readiness blockers.
Campaign editor	Trigger > Audience > Messages > Sending rules > Review. Draft autosave with version conflict handling.
Review / dry run	Eligible and excluded counts, reason breakdown, sample messages and first local/UTC dates. Zero live sends.
Campaign activity	Enrollment and step history with scheduled time, status, reason, template version and linked outcomes.
Campaign settings	Timezone, holidays, caps, global pause and read-only integration readiness.

The main action is **Activate campaign**, not Send to customer. A separately labelled test preview may use a simulator; a real test send requires an explicitly selected test recipient and must not create live campaign enrollment.

04 / FRONTEND CONTINUED

Campaign settings and validation

Setting	Example / allowed value	Required validation
Name and scope	3-100 characters; selected shop/branches	Tenant scope from auth; reject unauthorized branch/customer references.
Preset / trigger	Allowlisted trigger type	Require type-specific fields; reject unknown rule operators.
Timezone / send time	Asia/Kolkata; 10:00	Valid IANA zone and HH:mm. Store date-only occasions separately.
Lead or delay	Birthday -14 days; enquiry +2 days	Bounded integer; valid before/after semantics; no negative absolute delay.
Start/end and expiry	Activation window; per-step useful-until	A usable slot must exist. No send after benefit/event fulfillment deadline.
Allowed hours / blackouts	EXAMPLE 10:00-19:00; chosen dates	Start inclusive/end exclusive; v1 rejects overnight/equal hours. Blackout adjustment may not exceed expiry.
Leap-day rule	FEB_28 / MAR_01 / LEAP_ONLY	Explicit selection for recurring Feb 29 dates.
Steps and stop conditions	1-3 stable steps	Increasing occasion/event offsets; no loops. Required reply/purchase/opt-out policy.
Priority and caps	EXAMPLE 1/24h and 3/7d marketing	Tenant-wide across campaigns; owners may tighten inherited limits.
Resume / outage policy	Resume: FUTURE_ONLY; outage grace: 2h	Product defaults; no silent bulk catch-up.
Historical imports	No backlog by default	New imported customers may join future occurrences only.

Required settings notice: "This module runs auto campaigns using an existing WhatsApp connection and approved templates. WhatsApp setup and template approval are managed by the integration developer."

Do not add token entry, business verification, Create template or Submit for approval controls. The campaign developer must still enforce readiness at activation and execution.

05 / CONTENT CONFIGURATION

Message steps and template mapping

Field	Behaviour
stepId / step name	Stable identifier survives edits. Never use array index as identity.
Schedule	Offset from occasion local date or source event time. v1 does not support arbitrary conditional loops.
Prerequisite	NONE or PREVIOUS_ACCEPTED. A reminder can require evidence that its invitation was accepted.
Template	One existing approved template ID + exact locale + immutable schema/content revision.
Bindings	Allowlisted source paths or approved literals: customer name, shop, validated offer, event date, product or appointment.
Business validation	Price, credit, inventory, venue, dates and links come from current authoritative records.
Media / buttons	Only adapter-supported template components with authorized tenant-owned media and valid links.
Stop behaviour	Sales default stopOnReply=true (staff hold), stopOnGoal=true (skip). MARKETING_OPT_OUT cancels promotions; CHANNEL_SUPPRESSED or deletion cancels all pending steps.
Expiry	Explicit useful-until rule; never wait indefinitely for a cap, stock or template.

Preview and activation

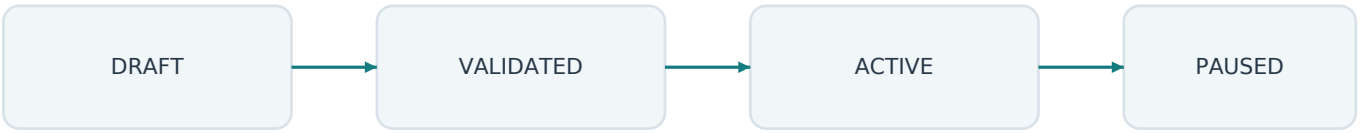
The server resolves bindings, checks types and renders the exact selected template. Missing required values produce a visible skip reason for affected customers. An unusable template, broken mapping source or missing connection blocks activation. An individual customer missing a birthday does not block the whole campaign.

Dry runs persist a report, draft fingerprint and sample eligibility evidence; they create no live execution. Activation must reference a dry run for the same draft fingerprint. Zero current matches is a warning: a future stock event or birthday may still qualify.

No silent language substitution, AI rewrite, invented offer, unresolved placeholder or automatic template switch. A changed template revision requires an explicit campaign revision. [M1]

06 / LIFECYCLE

Campaign lifecycle and published versions



Editing an active campaign
Create draft version N+1.
Published version N keeps running.

Publish revision
New enrollments use N+1.
Existing enrollments retain N.

ARCHIVED
No new enrollments; pending steps cancelled; history retained.
Resume uses future-only behaviour by default, with missed steps logged.

Version contract

Save edits to a draft with expectedVersion. Activation freezes an immutable campaignVersion and records activatedBy, activatedAt, reviewFingerprint and nextEvaluationAt. Editing an active campaign creates another draft; publishing affects only new enrollments by default.

Existing enrollments keep the trigger, sequence, audience snapshot and timezone they were created with. Live permission, customer deletion, stock, order state, global pause and template usability override every version.

End and archive

The configured endAt stops new enrollments and expires steps beyond the campaign deadline. Archive also cancels remaining unsubmitted work and preserves history. Neither operation recalls messages already accepted by the provider. Treat accepted and outcome-unknown executions conservatively.

07 / CUSTOMER SELECTION

Audience logic and eligibility

Select eligible customers automatically. Use a controlled rules language, not arbitrary MongoDB filters or JavaScript from React. Every query includes tenant and permitted branch scope.

```
{
  "all": [
    {"field": "customer.status", "op": "eq", "value": "ACTIVE"},
    {"field": "preferences.weave", "op": "in", "value": ["silk"]}
  ],
  "any": [
    {"field": "preferences.budgetMaxPaise", "op": "gte",
     "value": 600000},
    {"field": "tags", "op": "contains", "value": "occasionwear"}
  ],
  "none": [
    {"field": "serviceIssueOpen", "op": "eq", "value": true}
  ]
}
```

Semantics: all predicates must pass; any is optional but at least one must pass when present; none predicates must all fail. Missing fields fail ordinary comparisons. Empty any/none are neutral. An entirely empty audience is invalid unless the owner explicitly selects ALL_ELIGIBLE.

Mandatory gates outside the editable audience

- Valid, active customer and destination; marketing permission for promotional steps; channel suppression always respected.

- Required trigger data and business eligibility, such as a valid occasion date or unresolved enquiry.

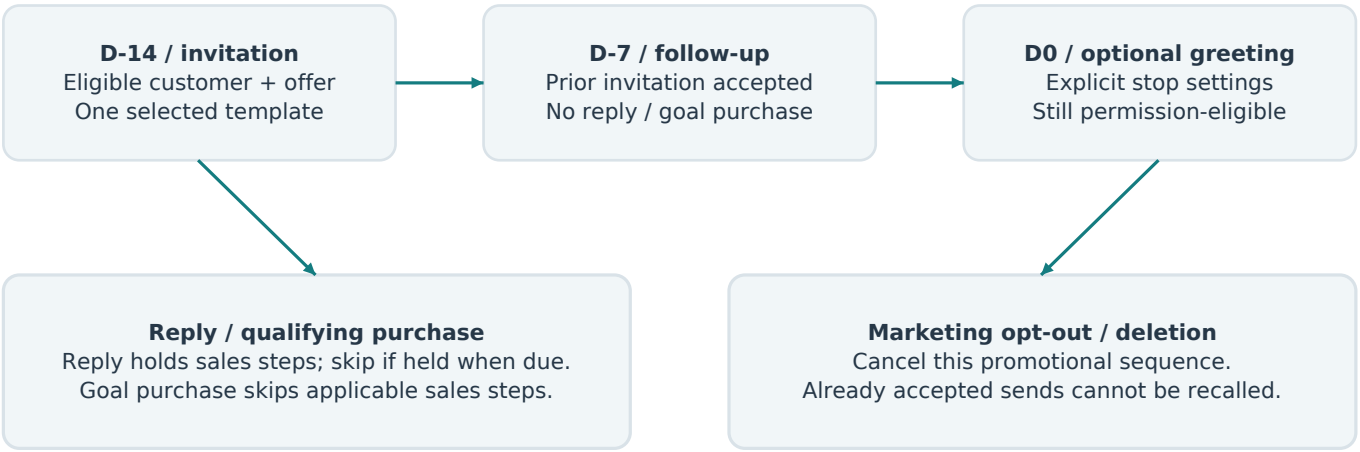
- No duplicate enrollment for the same occurrence; no duplicate destination contact when household records overlap.

- Permission and current business conditions are checked again when each step becomes due.

Store a concise match explanation, such as "birthday in 14 days; opted in; silk preference". Do not infer religion, relationship status, birthday or partner contact details. Unsupported fields/operators fail validation rather than broadening the audience.

08 / OCCASION SCENARIOS

Birthday campaign: complete behaviour



The occurrence is the birthday year, not the send year.
Every step has its own stable ID and one execution record.

Required data and planning

Read customer birthday month/day, recipient identity, permission and campaign timezone. Resolve the next annual occurrence using the explicit leap-day rule. EXAMPLE preset: invitation D-14 at 10:00; optional reminder D-7; optional greeting D0. Each step selects its own existing approved template.

Eligibility and stop rules

Recommended goal: an order explicitly tagged to this occasion or a staff-recorded shopping-complete event. An optional recent-purchase rule needs an explicit window; it must not claim every purchase was for the birthday. A goal skips applicable sales steps. A reply holds them for staff handling: if still held when due, skip that step. Explicit hold resolution allows only future steps to be re-evaluated. An optional greeting follows its own stop settings; permission and caps always apply.

Occurrence and idempotency

Occurrence ID: **birthday:2027**. A January 2027 invitation sent in December 2026 still uses 2027. One enrollment per campaign/customer/occurrence and one execution per stable step. Changing a template or birthday cannot automatically resend a completed annual step.

08 / OCCASION SCENARIOS CONTINUED

Anniversary, gifting and wedding campaigns

Scenario	Plan and inputs	Stop / edge cases
Anniversary	Voluntarily saved eventId + month/day. EXAMPLE invitation D-21, reminder D-7, greeting D0. Goal window and template chosen by owner.	Removed date, opt-out, goal purchase, reply or expired benefit suppress the relevant steps. Distinguish multiple saved occasions by eventId.
Gift reminder	Reminder belongs to the customer who requested it. Save date, recipient label if volunteered, budget and selected template.	Do not discover/message a partner. Gift bought or reminder deleted ends that occurrence.
Wedding / family event	Exact event date, requested categories, budget and actual tailoring/dispatch lead time. Example first invite D-45 when the shop needs that preparation time.	Cancelled/postponed event, completed shopping or insufficient production time blocks outdated messages.
Optional appointments	Offer a real shopping/fitting slot through a supplied booking link.	Do not claim an appointment is booked merely because the invitation was delivered.

Stable identities

Annual anniversary: anniversary:eventId:year. One-time wedding: wedding:eventId. Postponing a wedding revises the same event identity; it does not create a fresh campaign opportunity automatically. Recompute only unsubmitted steps, preserve accepted steps and log the old/new dates.

Useful-before deadline

A wedding invitation may expire before the wedding itself: latestUsefulAt = eventDate minus required fulfillment lead time. The planner must not move a blocked message into a slot after that deadline. If the event is too close, skip with INSUFFICIENT_LEAD_TIME and show the owner the reason.

Date calculation is owned by this campaign developer. The other developer only supplies usable templates and WhatsApp readiness; approval work is not part of these scenarios.

09 / MERCHANDISE SCENARIOS

Festivals, collections and stock matching

Scenario	Engine behaviour	Send-time conditions
Festival collection	Owner creates seasonId, actual date, collection and audience. Example launch D-21 and optional follow-up D-7.	Season/offer active; relevant sellable stock; recorded preferences. Never infer religion from names.
New collection	On collection.published, match customer preferences, budget and stated interests. One enrollment per customer/release.	At least one eligible item remains. Collection and approved message bindings still match.
Requested restock	On a genuine sellable stock transition, join open product requests to exact SKU/variant.	Request unresolved; exact colour/size/weave condition true; no relevant purchase or reservation conflict.
Similar handloom arrival	Use a separate matching rule and template family for similar pieces.	Never call a similar item the exact requested piece or imply a reservation.
Selected-stock offer	Owner publishes offerId with eligible items, actual prices/terms and useful-until date.	Matching stock and offer still valid; no fabricated scarcity or expired discount.

Matching and fan-out

Normalize preferences and product attributes into controlled values. Budget is integer minor units; size/colour/weave comparisons use recorded attributes. Store match reasons. One inventory receipt with several units must not send one message per unit. Consolidate multiple matches for a customer into one selected template payload when supported.

Occurrence keys

festival:seasonId; collection:releaseId; arrival:arrivalEventId:requestId; stockoffer:offerId. Restock event IDs must represent a true published availability change, not every stock recount. Replaying an event must not create new opportunities.

A sent message does not reserve a unique saree. Recheck availability at execution and again before submission, while acknowledging a sale can still occur immediately after the check.

10 / CUSTOMER LIFECYCLE SCENARIOS

Enquiry, repeat purchase and win-back

Preset	Trigger / example timing	Exit and edge cases
Unresolved enquiry	Real enquiry ID and configured last-interaction timestamp. Example first nudge +24h, second +72h from the same anchor.	Stop/hold on reply, staff takeover, resolved enquiry, purchase or unavailable item. A record edit is not customer activity.
Complete the outfit	After a qualifying completed order; example +5 days. Match compatible blouse, accessory or other stocked item.	Suppress returned relevant lines, incompatible/unavailable complements, or already-purchased complements.
First to second purchase	First qualifying order; example +30 days with one optional later step.	A second qualifying purchase stops the sequence. Full refund of the first invalidates its eligibility.
Win-back	Daily indexed evaluation of last qualifying purchase + chosen category threshold, e.g. 90 or 180 days.	A new purchase cancels old steps and resets the anchor. No purchase history is not a win-back audience.

Purchase definition

Default qualifying purchase requires a paid, completed order with positive non-refunded value. Pending/unpaid/cancelled orders do not count. Partial returns affect relevant lines and net amounts. The owning order service supplies authoritative status; this developer builds campaign aggregates and recomputation from those events.

Inactivity is anchored, not repeated daily

Store lastQualifyingOrderId, lastQualifyingPurchaseAt and aggregateVersion. Occurrence = winback:lastQualifyingOrderId:ruleId. Repeated daily scans reuse that occurrence. A new order resets it; a refund may restore an older anchor but must not resend an already completed occurrence.

Reply handling

A reply linked to a campaign message marks that enrollment. Without reliable message context, conservatively hold active sales sequences for that customer/store after a new inbound reply; do not claim campaign attribution. Staff can resolve the conversation hold, but missed or expired steps do not automatically replay.

11 / BOUTIQUE SCENARIOS

Boutique appointments and fulfillment

Campaign	Plan	Cancel / recompute rules
Fitting appointment	Confirmation after booking and optional reminder 24h before. Use real appointmentId, local time, branch and status.	Cancelled/completed appointment stops reminders. Reschedule moves unsubmitted reminders; already-sent confirmation may require a distinct update event.
Alteration milestone	Trigger on actual stage change. Choose a stage-specific approved template.	Do not imply the full outfit is ready when only a blouse or alteration stage is complete.
Outfit ready	On full ready-for-collection status; optional reminder after a configured uncollected interval.	Collected/delivered/cancelled/rework state stops pending notices. If delivery is arranged, do not send pickup instructions.
Delivery update	Trigger on verified dispatched/delivered events, with valid tracking reference where applicable.	Label creation alone is not dispatch. Ignore stale status versions and duplicate courier events.

Service purpose and matching

These are order-service sequences, separate from promotional campaigns. They use matching approved service templates and required order context. They do not consume the marketing cap, but still use per-order/step dedupe, permission policy, service contact spacing and quiet-hour settings.

Readiness versions

Occurrence examples: fitting:appointmentId; ready:orderId:readinessVersion; alteration:jobId:milestoneId. An order moving ready > rework > ready can have a new readiness version only for a genuine new event. A delayed old ready event must not override a current collected status.

Store the latest authoritative entity version and compare it during planning and dispatch. If part of an order is ready, use explicitly approved partial-ready wording or skip; never silently downgrade the customer promise.

12 / RETENTION SCENARIOS

Loyalty, credit, referrals and events

Preset	Engine inputs and timing	Required checks
VIP milestone	Authoritative program/milestone qualification event; one recognition per stable milestone.	Refunds may reverse eligibility before sending. Balance oscillations must not send repeated congratulations.
Credit expiry	Real creditId, positive balance, expiry and terms. Example reminder D-14 and D-3.	Redeemed/expired/extended credit updates pending steps. Display current balance; never invent expiry urgency.
Referral invitation	Completed order + active program/link; example +7 days.	Skip cancelled/refunded source orders or active complaints where policy suppresses marketing. Sending does not issue a reward.
Exhibition / trunk show	Published eventId, venue, date, capacity and chosen geography/interests. Example invitation D-14.	Cancelled/full event or existing RSVP can stop generic invitations. Attendee reminders are separately configured steps.

Stable occurrence identities

vip:programId:milestoneId; credit:creditId; referral:orderId:programId; storeevent:eventId. Extending a credit expiry revises the same occurrence. A program revision alone must not regenerate previously sent recognition.

Benefit ownership

The campaign engine may reference a valid discount, store credit, appointment or referral benefit created by its owning module. It does not fabricate reward balances, award referral credit on message delivery, or use a customer review as an automatic condition for a reward.

Owner explanations

Show practical reasons: "Skipped: credit already redeemed", "Scheduled: event reminder tomorrow", "Stopped: customer already registered". These are business outcomes, not generic sending errors.

13 / SCHEDULING ALGORITHM

Scheduler, local time and recovery

Three durable evaluation paths

Path	Algorithm	Checkpoint
Calendar planner	At activation and periodic scans, resolve upcoming occurrences and upsert enrollments/steps within the planning horizon.	Campaign/version + scan cursor + completedAt.
Domain-event consumer	Persist event; validate source/version; evaluate matching active campaigns; create or reconcile enrollments and steps.	Unique producer event key + processed marker.
Inactivity evaluator	Scan an indexed nextEvaluationAt; verify latest order aggregate; enroll the due inactivity occurrence once.	Customer/category aggregate version and nextEvaluationAt.

EXAMPLE operations: plan every 15 minutes; claim due steps every minute; reconcile data nightly. Discover occasion dates from today minus the largest positive step offset through today plus the largest lead time plus 30 days. This includes future D+7 steps for a recently passed occasion. Reconcile existing active enrollments independently of that window. Never use one setTimeout per future message.

For an existing enrollment, load its pinned campaignVersionId before upserting its stable steps. Never add steps from the newest active version to an older enrollment. Use bounded cursor batches and durable checkpoints.

Time conversion

For a date trigger: resolve occurrence local date, add signed calendar-day offset, apply the step local time, then convert with the campaign IANA timezone to UTC. Store original local date/time/zone and resolved dueAt. For an event delay, add elapsed duration to authoritative occurredAt, then fit the result into permitted local sending hours.

DST rule: shift a nonexistent local time to the next valid instant; choose the first occurrence of a repeated local time. Feb 29 uses the selected annual policy. Date-only birthdays must never be converted through midnight UTC.

Late work

Normal activation/import and explicit resume use FUTURE_ONLY: elapsed steps are logged as skipped. EXAMPLE technical-outage grace is 2 hours, bounded by expiresAt, send hours and current relevance. Longer outages skip obsolete steps; never replay a backlog in a burst. A future optional greeting can still run after a missed invitation.

Set expiresAt to the earliest applicable step lateness cutoff, campaign end and source latestUsefulAt. Moving a send into allowed hours cannot extend that deadline. A staff conversation hold is not a technical outage and does not earn catch-up grace.

14 / DEDUPLICATION + CONTACT POLICY

Duplicate prevention and campaign collisions

```
enrollmentKey = tenantId + campaignId + customerId + occurrenceId
executionKey  = enrollmentId + stableStepId
deliveryKey   = executionId
// Do not include campaign version, template revision, retry count,
// worker ID or schedule time in these business-identity keys.
```

One record per business step

Use unique indexes and transaction-safe upserts. Repeated scans, webhook replay and publish operations reach the same enrollment/execution. Date corrections retain identity unless the occurrence year changes; use the correction rules on page 20. A new published version never makes an executed annual step eligible again.

Deterministic collision resolution

Before dispatch, lock/arbitrate by tenant + canonical destination. Among currently due eligible marketing steps choose highest priority, then earliest expiry, then stable creation time/ID. EXAMPLE priorities: birthday/anniversary 80, wedding 70, requested stock 60, festival 50, win-back 30. The owner can configure priorities within an allowlisted range.

EXAMPLE default caps: one marketing contact per rolling 24 hours and three per rolling 7 days. Count reservations, accepted and uncertain submissions across all campaigns and duplicate customer records sharing a destination. The losing step is deferred to the next available slot only if it remains useful; otherwise record SKIPPED_CONTACT_CAP.

Keep one rolling contact-ledger entry per execution, timestamped at reservation. Count entries in (now-window, now]; accepted/unknown entries age out normally, while unresolved submissions also retain their in-flight lock. Only proven non-acceptance releases that attempt's reservation. A safe retry reserves again at the new attempt time under the same execution key, without double counting. Lease expiry alone never releases an uncertain contact.

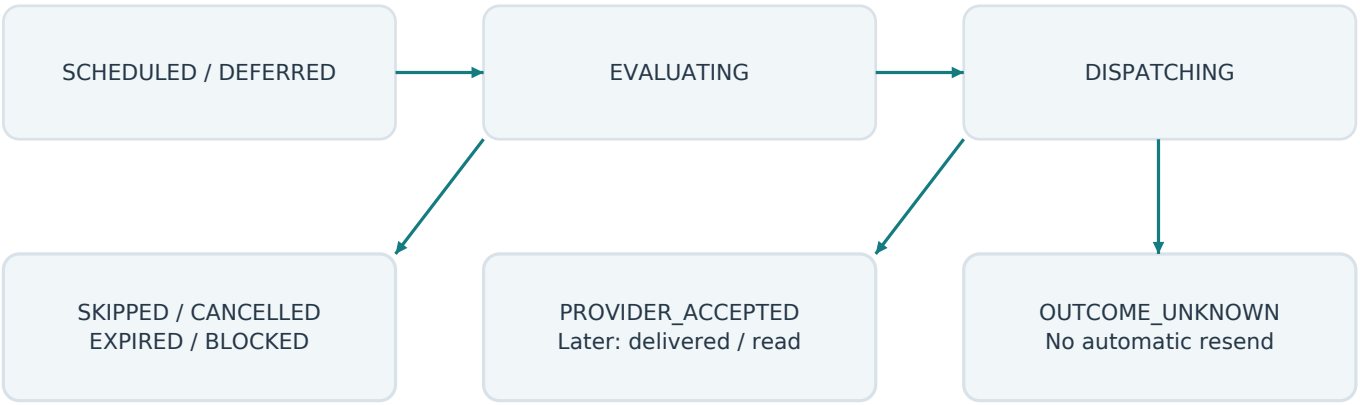
Boundaries

Never combine templates to solve a collision. Never assume a later-arriving high-priority event can undo a message already submitted. Service steps have separate configured spacing and entity-level dedupe; a marketing template cannot bypass caps by claiming a service purpose.

Retain occurrence/execution identity protection across data cleanup for the campaign replay horizon. Cleanup must not delete the only evidence that a birthday step was already sent.

15 / EXECUTION ENGINE

Execution, stop rules and progression



Safe rejection
Bounded retry if still eligible and unexpired.
Same execution and delivery key.

Campaign progression
Each future step checks its prerequisite.
Acceptance is not a sale.

Evaluation when due

Claim one due step with a lease. Load its published rules and current customer, source, offer, stock, reply/goal state, template and connection. Check pause generation, date-plan revision, expiry, prerequisites, permission and contact capacity. Record a precise skip, block or deferral reason for every rejected candidate.

Resolve late, then freeze

Planned steps hold bindings and source references, not message bodies frozen weeks ahead. Resolve current values only when dispatch becomes due. Atomically reserve contact capacity and create the immutable delivery intent for the execution. Freeze recipient version, exact template revision, parameters and payload hash at that point.

Continue only from evidence

PREVIOUS_ACCEPTED checks the prior execution acceptance. Unknown or failed prior submission never counts as accepted. Delivery/read receipts do not count as replies or sales. Future independent steps may proceed only if their own rules allow it. An enrollment completes when all its steps have terminal business outcomes.

Recovery and wake-up rules

A reply hold is an enrollment flag, not a new execution state. BLOCKED_TEMPLATE wakes on readiness change or a bounded recovery scan. DEFERRED with WAITING_PREREQUISITE wakes on prior acceptance or its dueAt; skip at deadline if unmet. DEFERRED with RETRY_WAIT becomes due at retryAt and reuses its frozen intent. BLOCKED_STALE_INTENT stays blocked until expiry or explicit cancellation; never rewrite it. FAILED is terminal after permanent rejection or exhausted safe retries.

16 / CHANGE SEMANTICS

Pause, edit, source changes and resume

Action / change	Required engine behaviour
Pause campaign / global pause	Stop new enrollment and new submissions. Increment pauseGeneration. Retain schedules and show in-progress submissions separately.
Resume	Preview affected counts; default FUTURE_ONLY. Recheck future work and skip missed/expired steps. No automatic message burst.
Publish campaign revision	New enrollments use new version; existing enrollments retain old rules. Show old-version enrollment counts.
Change birthday / occasion date	Same occurrence year: revise pending times and planRevision. Across years: cancel the obsolete unsent plan and evaluate the corrected year under FUTURE_ONLY. Preserve completed/unknown evidence; see correction note below.
Reschedule appointment	Move pending reminders; preserve sent confirmation. An explicit appointment-change event may generate a distinct update message.
New customer destination	Before dispatch, use only current verified identity and permission for that destination. After intent freeze, a change blocks that intent; never reroute it.
Preference change / deletion	MARKETING_OPT_OUT cancels promotional steps. CHANNEL_SUPPRESSED or deletion cancels all pending steps. Recheck at dispatch; later opt-in never resurrects cancelled occurrences.
Goal purchase / staff completion	Apply each step's stopOnGoal. Optional greeting follows its separate rule; never bypass channel permission.
Reply / staff handling	stopOnReply=true creates an enrollment hold. Keep future steps scheduled; at due time, skip if still held. Staff resolution re-evaluates future steps only.
Template paused / removed	Block affected steps with fix owner. Once restored, re-evaluate only unchanged useful work; no silent replacement.

Across-year correction: link old/new enrollments with a correctionGroupId and retain the old pinned version when creating the replacement. Existing target enrollment keeps its version. In that correction group, accepted/unknown stable steps suppress replacement copies. Preserve old evidence and never attach a December 2026 date to birthday:2027. Ordinary later annual occurrences are not part of this correction group.

Replan only executions with deliveryIntentId=null. Once an intent exists, changed date/price/destination or relevant source bindings block it as stale, even after proven rejection; safe retries retain the exact frozen payload. Atomically update enrollment planRevision and affected execution revisions/schedules. Final submission must compare the same execution revision, so a racing replan invalidates the claim.

V1 does not migrate active enrollments to a new campaign version automatically. If the owner needs different content for existing customers, pause the old campaign and use an explicit reviewed future enhancement; changing stepId to force a resend is not permitted.

17 / DATA MODEL

MongoDB collections and relationships

Collection	Responsibility / key fields
campaigns	Stable campaign ID, tenantId, status, activeVersionId, draftVersion, pauseGeneration, activation/end timestamps.
campaignVersions	Immutable trigger, audience, steps, timing, contact policy, goal/stop rules, publishedBy/At and review fingerprint.
campaignEnrollments	Campaign/version/customer/occurrence identity; source reference/version; status, reply hold, goal state, planRevision.
campaignExecutions	One step per enrollment: dueAt, expiresAt, local date/zone, state, prerequisite, lease, planRevision and deliveryIntentId.
campaignEventInbox	Producer event identity, tenant, entity ID/version, occurredAt, receivedAt, schema version and processing status.
campaignCheckpoints	Leased planner/event/inactivity scan cursors and evaluation health.
campaignCustomerFacts	Rebuildable order/interest aggregates, last qualifying purchase, nextEvaluationAt and authoritative source versions.
campaignContactGates	Destination-scoped mutex/version and bounded rolling contact reservations/accepted/unknown entries.
campaignDeliveryIntents	Immutable single-message payload, execution key, attempt status, provider ID and delivery evidence.
campaignGoalEvents / audit	Explicit staff/domain goal evidence and append-only configuration/operational history.
campaignDryRuns	Draft fingerprint, counts/reasons, sample plans, evaluatedAt and source snapshot metadata.

Use BSON Date UTC instants, integer minor-unit money and explicit date-only objects. Tenant scope belongs on every collection. Keep message/phone data encrypted or protected; retain masked projections for routine history. The template/connection catalogue is a read-only integration dependency.

17 / DATA MODEL - BIRTHDAY PRESET

Campaign configuration example

```
{
  "campaignId": "cmp_birthday", "tenantId": "shop_a",
  "name": "Birthday shopping", "version": 1,
  "industry": "BOUTIQUE", "purpose": "MARKETING",
  "trigger": {
    "type": "ANNUAL_DATE", "field": "customer.birthday",
    "leapDayPolicy": "FEB_28"
  },
  "timezone": "Asia/Kolkata",
  "audience": {"mode": "ALL_ELIGIBLE"},
  "goal": {"type": "OCCASION_TAGGED_OR_STAFF_COMPLETED"},
  "steps": [
    {
      "stepId": "invitation", "offsetDays": -14,
      "localTime": "10:00", "prerequisite": "NONE",
      "templateRef": {"id": "tpl_birthday_invite", "revision": 3,
                      "language": "en"},
      "bindings": {"name": "customer.firstName",
                   "shop": "shop.displayName"},
      "stopOnReply": true, "stopOnGoal": true,
      "maxLatenessMinutes": 120
    },
    {
      "stepId": "followup", "offsetDays": -7,
      "localTime": "10:00", "prerequisite": "PREVIOUS_ACCEPTED",
      "templateRef": {"id": "tpl_birthday_followup", "revision": 1,
                      "language": "en"},
      "bindings": {"name": "customer.firstName"},
      "stopOnReply": true, "stopOnGoal": true,
      "maxLatenessMinutes": 120
    }
  ],
  "priority": 80,
  "resumePolicy": "FUTURE_ONLY",
  "importPolicy": "FUTURE_ONLY",
  "enabledBranches": ["branch_main"]
}
```

Illustrative IDs and template names; they are not approval claims. A third greeting step is optional. Its stop flags, template and deadline must be configured explicitly. Effective store-wide caps, send hours and permission rules remain mandatory even when not repeated in this version object.

17 / DATA MODEL - EXECUTION RECORDS

Enrollment, planned step and delivery data

```
// campaignEnrollments
{
  "id": "enr_001", "tenantId": "shop_a",
  "campaignId": "cmp_birthday", "campaignVersionId": "cv_001",
  "customerId": "cus_001", "occurrenceId": "birthday:2027",
  "sourceRef": {"type": "CUSTOMER_DATE", "id": "cus_001",
                "version": 8},
  "occasionLocalDate": "2027-01-05", "planRevision": 1,
  "status": "ACTIVE", "replyHoldAt": null, "goalReachedAt": null
}
// campaignExecutions: planned, no rendered message yet
{
  "id": "exe_001", "tenantId": "shop_a",
  "enrollmentId": "enr_001", "stepId": "invitation",
  "dueAt": "2026-12-22T04:30:00Z",
  "expiresAt": "2026-12-22T06:30:00Z",
  "localDate": "2026-12-22", "localTime": "10:00",
  "timezone": "Asia/Kolkata", "planRevision": 1,
  "state": "SCHEDULED", "deliveryIntentId": null
}
// campaignDeliveryIntents: materialized only at dispatch
{
  "id": "di_001", "tenantId": "shop_a", "executionId": "exe_001",
  "deliveryKey": "exe_001", "connectionId": "conn_shop_a",
  "recipientVersion": 4, "encryptedRecipient": "<encrypted>",
  "templateRef": {"id": "tpl_birthday_invite", "revision": 3,
                  "language": "en"},
  "parameters": {"name": "Lakshmi", "shop": "Sample Boutique"},
  "payloadHash": "<canonical-hash>", "state": "SUBMITTING",
  "attemptId": "attempt_001", "providerMessageId": null
}
```

10:00 Asia/Kolkata equals 04:30 UTC. Execution identity does not change when the date plan is corrected. Each attempt has its own ID, while retries preserve the same immutable delivery intent and business step identity.

18 / INPUT EVENTS

Domain events and source-data contracts

```
{
  "schemaVersion": 1, "producer": "orders",
  "eventId": "evt_101", "tenantId": "shop_a",
  "type": "order.completed", "entityId": "ord_001",
  "entityVersion": 6,
  "occurredAt": "2026-12-23T08:00:00Z",
  "receivedAt": "2026-12-23T08:00:02Z",
  "customerId": "cus_001",
  "data": {
    "paid": true, "status": "COMPLETED", "netPaise": 580000,
    "occasionRef": "birthday:2027",
    "lineRefs": ["line_001"]
  }
}
```

Source	Minimum contract
Customer / permission	Stable customer, verified destination/version, month/day dates, preferences, consent/suppression and data-change version.
Orders / refunds	Paid/completed/cancelled/refunded states, effective net amounts, line/product IDs, occasion/campaign references when recorded.
Inventory / collection	Published event identity, exact variants, sellable stock, prices and latest source version.
Appointment / alteration	Actual stage, branch, local date/time, readiness/reschedule versions and cancellation/collection state.
Reply / preference events	Authenticated connection/customer mapping, inbound time and message context when present; opt-out updates.

Persist each event before acknowledging. Process its effects and processed marker atomically or through a recoverable outbox. Event payload is a hint; fetch current authoritative state before scheduling sensitive claims. An older event cannot restore a state superseded by a newer entity version.

A stale timestamp, unavailable source or incompatible schema produces a visible processing error/quarantine state. Do not silently treat missing data as eligible.

19 / MONGODB IMPLEMENTATION

Indexes, transactions and worker claims

```
db.campaignVersions.createIndex(
  {tenantId: 1, campaignId: 1, version: 1}, {unique: true});
db.campaignEnrollments.createIndex(
  {tenantId: 1, campaignId: 1, customerId: 1, occurrenceId: 1},
  {unique: true});
db.campaignExecutions.createIndex(
  {tenantId: 1, enrollmentId: 1, stepId: 1}, {unique: true});
db.campaignExecutions.createIndex(
  {state: 1, dueAt: 1, _id: 1});
db.campaignEventInbox.createIndex(
  {tenantId: 1, producer: 1, eventId: 1}, {unique: true});
db.campaignDeliveryIntents.createIndex(
  {tenantId: 1, executionId: 1}, {unique: true});
db.campaignContactGates.createIndex(
  {tenantId: 1, destinationHash: 1}, {unique: true});
```

Transactional boundaries

1. Publish campaign version and change activeVersionId atomically. 2. For large event fan-outs, commit each bounded batch's enrollment/step effects with its cursor; mark the event processed only after the final batch. Replay resumes the cursor. 3. At submission, validate control generations, reserve capacity, freeze one intent and mark DISPATCHING atomically. Never perform provider HTTP inside a transaction callback. [D1, D2]

Claims and control generations

Claim due SCHEDULED/DEFERRED work with findOneAndUpdate into EVALUATING using an expiring unique lease token. Final submission CAS must match token, expected planRevision, current state and lease.until > serverNow, with expiresAt > serverNow. A stale worker must fail. Reclaim only expired pre-submission leases. [D3]

Use a small campaign control-gate document for pauseGeneration and a tenant control gate for global pause. Submission transactions conditionally update the expected gate versions; pause changes those same documents. A snapshot read alone is insufficient to serialize pause against submission. Lock/update tenant, campaign and destination gates in a consistent order.

Preserve unique enrollment/execution tombstones before pruning detailed records. Tombstone creation and detail removal must be atomic, and re-enrollment must check both live and retained keys. Annual replay protection lasts for the campaign lifetime; retention beyond that is an explicit product/data policy.

20 / APPLICATION CONTRACTS

Node API surface and permissions

All routes are application APIs under /api/v1. Use authenticated tenant context and role checks. Route IDs never grant cross-tenant access. Mutations require idempotency keys; edits/publish additionally require expectedVersion.

Endpoint	Responsibility / result
GET /campaign-presets	Supported presets, required data and implemented capabilities.
GET/POST /campaigns	List campaigns or create a draft. No sends on create.
GET /campaigns/:id	Draft, active version, status, health and old-version enrollment counts.
PATCH /campaigns/:id/draft	Update allowlisted configuration; 409 on stale version.
POST /campaigns/:id/validate	Structured errors/warnings with fieldPath and resolutionOwner.
POST /campaigns/:id/dry-runs	Persist a read-only evaluation report; never create live jobs.
POST /campaigns/:id/activate	Publish a valid reviewed version and start planning. Returns next evaluation, not "sent".
POST /campaigns/:id/publish-revision	New immutable version; existing enrollments remain pinned.
POST /campaigns/:id/pause|resume|archive	Separate routes/actions with reviewed affected counts and defined lifecycle rules.
GET /campaigns/:id/enrollments|history	Separate paginated views; customer, occurrence, step, schedule, outcome and reason.
POST /enrollments/:id/resolve-hold|mark-goal	Separate audited staff actions; retain cause and supporting record.
GET/PATCH /campaign-settings	Effective timezone policy, send hours, blackouts, caps and global pause.
GET /approved-template-catalogue	Read-only integration catalogue; exact supported locale and revision.
POST /internal/campaign-events	Authenticated domain-event ingestion with durable dedupe.

Shop staff may view activity and resolve permitted customer holds. Owner/admin roles manage campaigns and global policy. Internal event producers are scoped by tenant and event family. Provider delivery-status ingress is implemented or connected by this developer as described next.

21 / WHATSAPP BOUNDARY

Message delivery: a supporting adapter

This is the delivery stage of the campaign engine. The other developer supplies the approved template catalogue, provisioned connection, server-side secret reference and account/template status changes. This developer implements dispatch and delivery-status processing, using an existing verified webhook router if available.

```
sendCampaignStep({
  executionId: "exe_001", deliveryKey: "exe_001",
  connectionId: "conn_shop_a", customerId: "cus_001",
  recipientVersion: 4,
  template: { id: "tpl_birthday_invite", revision: 3, language: "en" },
  parameters: { name: "Lakshmi", shop: "Sample Boutique" },
  expiresAt: "2026-12-22T06:30:00Z"
});
```

Concern	Required implementation
One selected message	One exact approved template per execution; bound variables only. No approval creation, automatic rewriting or SMS/email fallback.
Final checks	Campaign/global pause, source revision, permission, destination, template, contact reservation and expiry immediately before POST.
Success semantics	Persist provider message ID as accepted evidence. Later status events establish sent/delivered/read. Acceptance is not a sale.
Ambiguous timeout	Mark OUTCOME_UNKNOWN; do not automatically submit another copy. Correlate late evidence to the original attempt.
Safe rejection	Only documented non-acceptance can retry, with bounded backoff, same intent/key and all gates rechecked.
Event ingestion	Verify raw-byte provider signature or authenticated relay; durably store before acknowledgement; dedupe per status event, not only message ID.
Status ordering	Late sent cannot overwrite read/delivered. Missing read evidence is not unread. A read event implies delivery without supplying an exact earlier delivery timestamp.

After transaction commit, recheck the dispatch deadline before POST. A sweeper moves stale DISPATCHING/SUBMITTING attempts to OUTCOME_UNKNOWN after the bounded request deadline; it never requeues them. Correlate late HTTP/webhook results by immutable attempt ID/provider ID even after the preparation lease expires. Reconcile only that attempt; late evidence cannot initiate a new send.

Use server-side credentials and a pinned provider version. Current payload and event details come from official Meta references [M1-M4]. The user's phrase "direct sending" does not select Meta's separately named Direct Send capability; this release consumes approved campaign templates.

22 / OWNER RESULTS

Campaign reporting and explainability

Metric / view	Precise meaning
Eligible customers	Customers matching the evaluated rules at a stated time. A changing estimate, not a guaranteed send count.
Enrollments	Unique customer/occurrence records; multiple steps do not multiply the customer count.
Scheduled / deferred	Pending executions, with next date and why any delay occurred.
Skipped / cancelled / expired	Business-rule outcomes with reasons; not provider failures.
Accepted / delivered / read	Separate evidence counts per execution. Do not show accepted as delivered.
Replies / holds	Recorded inbound evidence and applied hold rules; distinguish exact reply context from conservative customer-wide hold.
Goal completions	Explicit tagged purchase or staff/domain completion, with supporting record and configured rule.
Linked orders / net sales	Orders with recorded campaign evidence under a declared attribution rule; cancellations/refunds reduce net value.
Customer timeline	Why selected, occurrence, planned/actual dates, template revision, status events and stop reasons.

Do not overstate sales impact

Delivery, greeting or store visit cannot be assumed to be a sale. Prefer order campaign links, referral/redemption codes or explicit staff links. If a timing-based model is used, label it as an attribution assumption. Never count the same order in several campaigns when reporting a single attributed-sales total; define precedence and show unassigned orders separately.

Sample owner explanations

"Selected because birthday is in 14 days." "Follow-up stopped because a reply is being handled." "Festival message deferred because another marketing campaign already contacted this customer." "Outfit reminder skipped because the order was collected."

Protect customer data in exports and logs. Log IDs, rule reasons, versions and timing by default; ordinary logs do not need full message bodies or phone numbers.

23 / REQUIRED BRANCHES

Edge-case decision register

Situation	Decision the engine must implement
Missing date / preference / language	Skip affected customer with reason; block campaign only if the required data source or template configuration itself is unusable.
Invalid date / Feb 29	Validate month/day; apply explicit non-leap-year policy. Never invent a date.
New activation during occasion cycle	Skip elapsed steps, schedule valid future ones. Independent greeting need not depend on a missed invitation.
Duplicate household phone	Keep customer identities separate; enforce destination-level caps and avoid conflicting simultaneous messages.
Customer deletion or withdrawn permission	Cancel pending affected steps; do not recreate records from replayed old events.
Source state changes after planning	Recompute unsent schedule or stop by current state; never trust the original audience snapshot alone.
Price or recipient changes during safe retry	Block frozen intent as stale; do not mutate the payload under its existing delivery key.
Busy cap / blackout / closed hours	Move only within useful window; otherwise skip. Record both original due time and adjusted time.
First step failed / unknown	A dependent reminder does not advance. Independent steps still obey current eligibility and caps.
Provider outage / worker restart	Durable jobs survive. Grace/expiry controls recovery; unknown submissions do not requeue.
Click pause while sending	Unsubmitted work stops; already-dispatching work may still send. UI reports that boundary honestly.
Campaign version edited mid-cycle	Old enrollment retains version. Completed logical steps never regenerate from a new revision.
Unknown event schema or stale entity event	Quarantine/record reason; never broaden audience or roll current state backwards.

No marketing inference should depend on a customer name, presumed relationship or guessed celebration. Eligibility must come from volunteered customer data or actual business records. The developer must expose skipped-reason counts so missing data is visible to owners.

24 / QA MATRIX A

Acceptance tests: campaign behaviour

ID	Scenario	Expected result
A01	Create birthday preset	Trigger, audience, steps and timing editor; no manual recipient selection required.
A02	Activate reviewed valid draft	Immutable version published; planner runs; no direct approval operation.
A03	Draft changed after dry run	Activation rejects old review fingerprint.
A04	No eligible customer today	Warning only when data source is usable; future customers can qualify.
A05	Missing template or mapping source	Activation blocked with field and owner of the fix.
A06	Birthday invitation due D-14	One customer/annual occurrence and selected template scheduled.
A07	January birthday invitation in December	Occurrence year is upcoming birthday year.
A08	Feb 29 in non-leap year	Configured Feb 28 / Mar 1 / leap-only behaviour, once.
A09	Anniversary record removed	All pending steps for that saved occasion stop.
A10	Wedding postponed after invitation	No repeat invitation; unsent relevant steps replan under same event ID.
A11	Customer replies after invitation	Sales steps held; if still held at due time, skip. Staff resolution permits future steps only.
A12	Goal purchase recorded	Applicable follow-ups stop; delivery alone never creates a goal.
A13	New purchase before win-back	Old enrollment stops; inactivity anchor resets.
A14	Returned complementary source item	Affected cross-sell is suppressed; unaffected lines follow defined rules.
A15	Requested exact size arrives	Only matching unresolved requests qualify.
A16	Unique saree sells before dispatch	No false availability message submitted.
A17	Outfit collected / fitting rescheduled	Obsolete reminders stop or replan; current details used.
A18	Credit redeemed / expiry extended	Outdated reminder suppressed or pending schedule recomputed.

24 / QA MATRIX B

Acceptance tests: scheduling and concurrency

ID	Scenario	Expected result
B01	Two planners evaluate same occurrence	One enrollment and one execution per stable step.
B02	Duplicate stock/order event replay	One logical event effect; no duplicate campaign contact.
B03	Consumer crashes before processed marker	Replay transaction/outbox finishes safely without duplicate enrollment.
B04	Date corrected within / across years	Correct year/key and future-only plan; correction lineage prevents repeats of accepted/unknown steps.
B05	Large historical customer import	Future-only planning; no catch-up burst.
B06	Worker downtime beyond grace	Expired/obsolete work skipped; useful future work remains.
B07	DST missing/repeated local hour	Next-valid / first-occurrence policy used once.
B08	Blackout moves send after useful deadline	Skip with reason instead of sending late.
B09	Birthday and festival compete	Deterministic priority; atomic cap; loser deferred or skipped.
B10	Two CRM records share destination	Destination cap enforced across records; identities not merged.
B11	Pause during evaluation	Control-gate generation invalidates stale submission.
B12	Pause after submission starts	UI says in progress/may send; no false recall.
B13	Resume with overdue jobs	Future-only rule; no burst of missed steps.
B14	Publish revision adding a third step	New enrollments use new version; older two-step enrollments remain two-step.
B15	Stale worker lease or planRevision	CAS fails; stale worker makes zero provider requests.
B16	Late old ready event after collection	Current entity version wins; no collection reminder.
B17	Refund restores previous inactivity anchor	Old completed occurrence is not re-enrolled.
B18	Cleanup followed by historical replay	Retained logical-key protection prevents duplicate step.
B19	Activate Jan 6 for Jan 5 occasion with D+7 step	Elapsed steps skipped; valid Jan 12 step discovered.
B20	Source replan races final submission	Execution-revision transaction/CAS chooses one order; stale claimant cannot send.

24 / QA MATRIX C

Acceptance tests: delivery and controls

ID	Scenario	Expected result
C01	Preview / dry run	Zero live message requests and zero live execution jobs.
C02	Template paused after activation	Affected step blocked with integration-owner reason.
C03	Opt-out after scheduling	Pending promotion cancelled; final gate prevents submission.
C04	Changed destination permission	No rerouting of frozen message; current identity/permission required.
C05	Missing variable / locale / invalid media	No partial or substituted message; precise reason.
C06	Provider accepted, response lost	Outcome unknown; no automatic second submission.
C07	Crash after DISPATCHING before HTTP	Unknown until proven otherwise; no blind requeue.
C08	Proven throttle rejection	Bounded retry, same key/payload; reservation released/reacquired once, all gates rechecked.
C09	Expiry reached during retry delay	Expired; no late offer.
C10	Webhook repeated / read before delivered	Idempotent evidence; no regression and no new send.
C11	Forged event / wrong tenant	No mutation or cross-tenant data exposure.
C12	Status event before response persistence	Durably retained and correlated to original attempt.
C13	Dependent reminder after unknown invitation	No advancement without acceptance evidence.
C14	Service update labelled as marketing escape	Registry/source-purpose rules prevent cap bypass.
C15	React refresh / duplicate activation click	Stable idempotency returns same operation; no duplicate publish.
C16	Conflicting settings/draft edit	409 version conflict; no silent overwrite.
C17	Delivered message without order evidence	No sale/revenue fabricated.
C18	Order refund / multiple campaign links	Net sales adjusted and single-attribution policy enforced.
C19	Template readiness restored	Useful blocked work re-evaluated; expired work remains unsent.
C20	Frozen intent source changes before safe retry	Intent blocked as stale; no payload rewrite or new automatic copy.

These 58 scenarios define required automated/integration tests. Passing must assert database outcomes, scheduler behaviour and provider call counts, not only UI messages. Production sending also requires a provisioned test connection and real approved template smoke test.

25 / DEVELOPER WALKTHROUGH

Worked birthday campaign end to end

Synthetic customer: Lakshmi, birthday 5 January, opted in, shop timezone Asia/Kolkata. Campaign version 1 has invitation D-14, follow-up D-7 and an optional greeting D0. All selected templates are supplied by the integration developer.

When	Engine action	Stored evidence
15 Dec 2026: owner activates	Validate settings/templates; publish v1; calendar planner finds 2027 occurrence.	Campaign ACTIVE; enrollment birthday:2027; three stable step records.
22 Dec 2026 10:00 IST 04:30 UTC	Invitation due. Recheck customer, offer, caps and template; bind current fields; submit one message.	exe_invitation becomes accepted with provider ID; delivery evidence arrives separately.
23 Dec: customer replies	Record reply and create staff-handling hold; invitation history retained.	replyHoldAt set; future follow-up remains scheduled with a hold indicator.
24 Dec: occasion order completed	Match explicit birthday occasion reference or staff-marked goal.	goalReachedAt with order reference; sales follow-up remains stopped.
29 Dec: follow-up date	Evaluate current stops; do not send sales reminder.	SKIPPED_GOAL_REACHED (goal precedes reply hold); reply evidence remains visible.
5 Jan 2027: optional greeting	If stop flags allow and permission/caps/template remain valid, send greeting. Otherwise record the reason.	One greeting execution; no second invitation. Enrollment completes after all steps terminal.

Reason precedence

For a single displayed primary reason, evaluate: customer deletion/channel suppression; campaign/global stop; invalid source/date; expiry; configured goal/reply stop; content/readiness; prerequisite; contact cap. Store all relevant diagnostics, but use one stable primary reason for counts.

Variation: birthday date corrected

A correction within the 2027 occurrence replans only pending times under birthday:2027. If January 2027 is corrected to December 2026, cancel obsolete pending work and evaluate birthday:2026 under FUTURE_ONLY. Link the correction group and suppress any step already accepted/unknown in the original enrollment. Preserve history; never re-send an invitation merely because its date or template changed.

26 / DELIVERY PLAN

Implementation order and release checklist

Milestone	What this developer delivers	Acceptance evidence
1. Foundation	Typed campaign schema, presets, Mongo indexes, source adapters and template catalogue read interface.	Schema/index checks and realistic fixtures.
2. Reference campaign	React editor + dry run + publish; birthday and anniversary planning, multi-step stops and history.	End-to-end date/goal/reply tests.
3. Reliable execution	Durable scheduler, control gates, collision caps, one-message adapter and status consumer.	Concurrency, expiry, pause and ambiguity tests.
4. Other triggers	Festival, stock, enquiry, purchase/inactivity, loyalty and boutique service evaluators.	Each preset has its own scenario fixtures and stop conditions.
5. Pilot readiness	Reporting, reason counts, audit, global pause, recovery runbook and test-connection sends.	All acceptance tests plus controlled pilot evidence.

Integration handoff required from the other developer

Ready connection identity, server-side secret reference, pinned provider capability/version, approved template ID/name/locale/category/schema/revision and status changes. Agree how authenticated provider events reach this developer's consumer. Template/account approval timelines must not be confused with campaign-engine completion.

Operational safeguards

Monitor planner lag, due-step age, skipped reasons, event backlog, unknown send outcomes and template blocks. Restart workers without losing schedules. Alert on stalled checkpoints; use bounded retries and tenant-level fair processing. Before pilot, record the expected load and test queue latency under provisioned provider limits.

The implementation must demonstrate automatic customer selection and scheduled execution. A manual sender, a static mockup or a list of campaign ideas does not satisfy this delivery scope.

27 / SOURCE REGISTER

References and specification boundaries

This is a proposed product/engineering specification. Business rules and example defaults are design decisions for GroVastra; official sources below support the technical integration patterns. Checked during this session on 13 October 2026.

[M1] Meta: message API and template payloads
https://developers.facebook.com/documentation/business-messaging/whatsapp/reference/whatsapp-business-phone-number/message-api

[M2] Meta: message status webhooks
https://developers.facebook.com/documentation/business-messaging/whatsapp/webhooks/reference/messages/status

[M3] Meta: webhook endpoint and signature verification
https://developers.facebook.com/documentation/business-messaging/whatsapp/webhooks/create-webhook-endpoint/

[M4] Meta: error codes and failure categories
https://developers.facebook.com/documentation/business-messaging/whatsapp/support/error-codes

[D1] MongoDB: atomicity and transactions
https://www.mongodb.com/docs/manual/core/write-operations-atomicity/

[D2] MongoDB: unique indexes
https://www.mongodb.com/docs/manual/core/index-unique/

[D3] MongoDB: findOneAndUpdate
https://www.mongodb.com/docs/manual/reference/method/db.collection.findoneandupdate/

[R1] React: effects versus user-triggered actions
https://react.dev/learn/synchronizing-with-effects

[B1] Reelo: auto-campaign reference page supplied by the user
https://reelo.io/auto-campaigns

Scope reminder for the developer

Implement the **auto-campaign engine**: configuration, audience evaluation, triggers, schedules, step conditions, stops, execution and reporting. WhatsApp is its sole delivery channel in this release. Consume existing approved templates and provisioned account readiness; do not build approval workflows.

This v2 document supersedes the earlier WhatsApp-sender handoff for assigning the work. Handloom stores, boutique owners and apparel shops are the intended businesses. No other industry workflow is included.